

Section 5.5 u-substitution

Ex ① $\frac{d}{dx}(e^{7x}) = e^{7x} \cdot 7$ Chain Rule

② $\int e^{7x} dx = \frac{e^{7x}}{7} + C$ ← because of the Chain Rule

u-substitution is like the Chain Rule for integration.

Ex ② Evaluate $\int e^{7x} dx$ using u-substitution.

$$\text{let } u = 7x$$

$$\frac{du}{dx} = 7$$

$$du = 7 \cdot dx$$

$$\frac{1}{7} \cdot du = dx$$

$$\int e^{7x} dx \quad \begin{array}{l} \leftarrow \text{replace } 7x \text{ with } u \\ \leftarrow \text{replace } dx \text{ with } \frac{1}{7} \cdot du \end{array}$$

$$= \int e^u \cdot \frac{1}{7} du$$

$$= \frac{1}{7} \cdot \int e^u du$$

$$= \frac{1}{7} e^u + C \quad \begin{array}{l} \text{Lastly, replace} \\ u \text{ with } 7x \end{array}$$

$$= \frac{1}{7} e^{7x} + C$$

Ex ③ $\int \sin(x^2) \cdot x dx$

let $u = x^2$

$$\frac{du}{dx} = 2x$$

$$du = \underbrace{2x \cdot dx}_{\leftarrow \text{make this match the original problem}}$$

For this step you can only multiply or divide by constants.

$$\frac{1}{2} du = x dx$$

$$\int \sin(x^2) \cdot \underbrace{x dx}_{\text{replace } x^2 \text{ with } u} \leftarrow \text{replace } x dx \text{ with } \frac{1}{2} du$$

$$= \int \sin(u) \cdot \frac{1}{2} du$$

$$= -\frac{\cos(u)}{2} + C$$

$$= -\frac{\cos(x^2)}{2} + C$$

For u-substitution:

- 1) Choose u to be the inner function
- 2) Find du
- 3) Replace All x 's with u 's
- 4) Integrate
- 5) Replace All u 's with x 's

Ex ④ $\int x \sqrt{x-5} dx$

Let $u = x-5$ $\xrightarrow{\text{solve for } x}$ $x = u+5$
 $du = 1 \cdot dx$

$\int (u+5) \sqrt{u} \cdot du$ distribute $u^{1/2}$

$= \int (u^{3/2} + 5u^{1/2}) du$

$= \frac{2}{5} \cdot u^{5/2} + \frac{10}{3} u^{3/2} + C$

$= \frac{2}{5} (x-5)^{5/2} + \frac{10}{3} (x-5)^{3/2} + C$

Ex ⑤ $\int e^{\sin x} \cdot dx$

let $u = \sin x$

$du = \cos x \cdot dx$

↑ this does NOT show up in the problem, so $u = \sin x$ will NOT work.

Try something else for u .

$$\int e^{\sin x} dx$$

$$\text{let } u = e^{\sin x}$$

$$du = e^{\sin x} \cdot \underline{\cos x dx}$$

↑ NOT in the original problem, so this u will NOT work.

$$\int e^{\sin x} dx \text{ cannot be done yet}$$

$$\underline{\text{Ex}} \text{ (6)} \int \frac{x}{1+10x^2} dx$$

$$\text{let } u = 1+10x^2$$

$$du = 20x dx$$

$$\frac{1}{20} du = x dx \leftarrow \text{this matches the numerator!}$$

Factor out constants



$$\frac{1}{20} \int \frac{1}{u} du = \frac{1}{20} \ln u + C$$

$$= \frac{1}{20} \ln(1+10x^2) + C$$

Ex ⑦ $\int \sin^2 x \cdot \cos x \, dx$

let $u = \sin x$

$du = \cos x \cdot dx$

$$\int u^2 du = \frac{u^3}{3} + C$$

$$= \frac{\sin^3 x}{3} + C$$

Ex ⑧ $\int \sin^3 x \cdot dx$ factor $\sin^3 x$ into $\sin^2 x \cdot \sin x$

$= \int \sin^2 x \cdot \sin x \, dx$ use $\sin^2 x = 1 - \cos^2 x$

$$= \int (1 - \cos^2 x) \cdot \sin x \, dx$$

$$\text{let } u = \cos x$$

$$du = -\sin x dx$$

$$\text{Factor out} \quad -du = \sin x dx$$

$$-\int (1 - u^2) du$$

$$= -\left(u - \frac{u^3}{3}\right) + C$$

$$= -\cos x + \frac{\cos^3 x}{3} + C$$